

Chapter 5: Morphology of Flowering Plants

Revision Notes — High-yield summary for quick recall before exams

1. The Root System

- Tap root (dicots, from radicle) vs Fibrous root (monocots, from stem base)
- 4 regions: Root cap → Meristematic activity → Elongation → Maturation (root hairs)
- Modifications: storage roots (radish/turnip/*Trichosanthes*), prop roots (banyan), pneumatophores (mangroves)

2. The Stem

- Develops from plumule; bears nodes, internodes, terminal & axillary buds
- Underground: tuber (potato), rhizome (ginger), corm (*Colocasia*), bulb (onion)
- Aerial: tendrils (grapevine), thorns (*Citrus*), phylloclades (*Opuntia*)
- Subaerial: runner, stolon, offset, sucker — all aid vegetative propagation

3. The Leaf

- Parts: leaf base, petiole, lamina; reticulate (dicot) vs parallel (monocot) venation
- Simple vs compound (pinnately – Neem; palmately – Silk Cotton)
- Phyllotaxy: alternate, opposite, whorled
- Modifications: tendrils (pea), spines (cactus), pitchers (*Nepenthes*)

4. Inflorescence

- Racemose: main axis grows continuously, flowers acropetal (older at base)
- Cymose: main axis ends in a flower, flowers basipetal (older at tip)

5. The Flower

- 4 whorls: calyx, corolla (accessory) + androecium, gynoecium (reproductive)
- Symmetry: actinomorphic (radial) vs zygomorphic (bilateral)
- Position: hypogynous (superior ovary), perigynous (half-inferior), epigynous (inferior)
- Aestivation: valvate, twisted, imbricate, vexillary (pea family)
- Placentation: marginal (pea), axile (tomato), parietal (mustard), free central (*Dianthus*), basal (sunflower)

6. Fruits and Seeds

- Fruit = ripened ovary; pericarp = epicarp + mesocarp + endocarp
- Parthenocarpic fruit develops without fertilization (banana)
- Dicot seed: 2 cotyledons, often non-endospermous (bean, gram, pea)
- Monocot seed: endospermous, single cotyledon = scutellum, coleoptile (plumule) & coleorhiza (radicle)

7. Family Solanaceae (Potato Family)

- Floral formula: $K(5) C(5) A_5 G(2)$
- Gynoecium bicarpellary, syncarpous, superior, axile placentation
- Fruit: berry or capsule
- Economic importance: food (tomato, potato, brinjal), spice (chilli), medicine (*Belladonna*, *Ashwagandha*), fumigatory (tobacco), ornamental (*Petunia*)

